

Efficient Raw-Waveform Audio Deepfake Detection: A CLAD-Inspired RawNet Approach vs. Mel-Spectrogram Baselines

May Rozen

Rotem Melamed

November 2025

Abstract

Abstract

Audio deepfakes have rapidly advanced to the point where synthesized speech can emulate a target speaker with high perceptual realism, creating security and societal risks ranging from fraud to the bypassing of voice authentication. While recent detectors achieve strong in-distribution accuracy, their robustness to natural, low-cost manipulations (e.g., volume changes, fades or noise) remains limited. In this project, we construct a complete pipeline for robust audio deepfake detection. On the data side, we curate real speech from the **VCTK** corpus and synthesize fakes with a neural TTS system (**XTTS** via Coqui TTS), conditioning on target speaker references to imitate identity. On the modeling side, we implement and compare two distinct architectures. Our primary model is a **CLAD-style detector** based on the **RawNet encoder**, which learns directly from the raw audio waveform. This model achieves robustness by being trained directly on on-the-fly manipulated data (e.g., noise, fades, and volume scaling) using a standard **Cross-Entropy loss**. We compare this against a baseline 2D CNN model operating on Mel-spectrograms. Initially, we hypothesized that the raw-waveform architecture is essential for robustness. However, our experiments reveal that a **Data-Centric approach** is paramount: applying our augmentation pipeline allows the lightweight CNN to achieve competitive accuracy. Nevertheless, the **RawNet-based CLAD model demonstrates superior stability** (lower Equal Error Rate) and computational efficiency by avoiding spectrogram conversion overhead.

1 Introduction

Advances in neural speech synthesis and voice conversion have made highly realistic audio deepfakes accessible, scalable, and increasingly difficult to distinguish from authentic recordings. This evolution poses a significant threat to security systems that rely on voice biometrics and contributes to the spread of misinformation and fraud.

The core challenge tackled in this work is the lack of robustness in current Audio Deepfake Detection (ADD) systems. Although these systems often perform well on clean, in-domain data, their performance degrades significantly when subjected to common acoustic artifacts encountered in the real world, such as Playback and Re-Recording (PBAR) effects, noise injection, or compression. These manipulations are trivial to apply, but often render standard detectors ineffective. Furthermore, methods designed to enhance robustness (e.g., contrastive learning frameworks) frequently rely on complex, multi-stage training methodologies, making them cumbersome, time-consuming, and resource-intensive to implement.

To address this, we propose and implement a highly efficient and simplified robust detection pipeline utilizing the CLAD (Contrastive Learning-based Audio deepfake Detector) framework. Our approach specifically leverages the RawNet architecture, which operates directly on the raw audio waveform, to learn manipulation-invariant features in a direct, End-to-End manner. We rigorously test the resilience of the model against a variety of common acoustic manipulations, demonstrating a marked improvement in generalization and robustness compared to non-robust baselines, while reducing architectural complexity and development overhead.

2 Data Preparation and Augmentation

2.1 Dataset Curation: VCTK and XTTS Synthesis

Our dataset comprises both real and synthetic audio files, processed for model compatibility:

- **Real Speech (VCTK):** Authentic audio files were sampled from the VCTK corpus, known for its diverse representation of speakers of various sex, ages, and accents. These samples serve as the Real class and provide the necessary reference voices for the synthesis process.
- **Fake Speech Generation (XTTS):** Deepfakes were generated using the Coqui TTS XTTS model, a state-of-the-art, cross-language neural Text-to-Speech system. This process leverages the model's Voice Cloning capabilities:
 1. A pretrained neural TTS model (XTTS) is loaded.
 2. Text transcripts are provided to the model.
 3. A real speaker's voice recording from VCTK is used as the Reference Voice.

4. The neural network synthesizes new speech for the provided text, precisely mimicking the timbre and characteristics of the target speaker’s voice.

This advanced synthesis process creates highly realistic voice clones that represent a demanding challenge for the detector.

All audio was standardized to a 16 kHz sampling rate and segmented to a fixed input length of 64,600 samples (~ 4 seconds) to meet the input requirements of the RawNet encoder. In total, the curated dataset comprises 1,342 real utterances and 1,342 corresponding synthetic clones (2,684 utterances overall), obtained from four distinct speakers. The speakers were partitioned into two speakers for training, one for validation, and one held out for testing, with all splits (training, validation, and test) strictly speaker-disjoint. This ensures that the detector cannot rely on memorizing speaker identity as the primary cue for classification.

2.2 On-the-Fly Manipulation Strategy

To explicitly drive the model towards robustness and build manipulation-invariant representations, we utilize a sophisticated on-the-fly augmentation strategy during training. The goal is to simulate realistic acoustic distortions without altering the core ‘real’/‘fake’ characteristic of the audio.

- **Training Augmentation:** For the **RawNet/CLAD** model, the data loader (using the **FlexibleAudioDataset** with **robust_train** mode) applied a random subset of the following three distinct manipulations to the raw waveform of each sample in the mini-batch:
 1. **Volume Scaling:** Random adjustments to the audio amplitude (gain/loss).
 2. **Fades:** Introduction of linear, exponential, or sinusoidal fades at the start or end of the segment.
 3. **Noise Injection:** Addition of low-level environmental or white noise.

Critical Implementation Detail: In each training step, the code randomly selected and applied **two** of these augmentations, ensuring that the model saw a constantly varying acoustic domain without being overwhelmed by excessive distortion. **In stark contrast, the 2D CNN baseline was trained solely on clean Mel-spectrograms, adhering to the standard practice for non-robust audio deepfake detection.**

- **Evaluation Manipulations (Testing).** Both the RawNet/CLAD model and the 2D CNN baseline were tested on the exact same attacked evaluation set. This set was generated using the **eval_robust** configuration, which applies a fixed combination of strong manipulations. The evaluation corruptions were deliberately chosen to be more severe than the stochastic augmentations used during training. This comparative stress test was crucial to assessing generalization: the ability of the CLAD model to maintain its high detection accuracy despite having seen only lighter corruptions in

training, versus the expected failure of CNN due to its exposure only to clean data.

This dual strategy—stochastic, lighter augmentations for CLAD training and deterministic, stronger corruptions for testing on both models—validates the model’s capacity to maintain its high detection accuracy even under adverse real-world conditions.

3 Methodology

The experimental methodology is designed to enable a controlled, comparative analysis between two distinct deep learning architectures for Audio Deepfake Detection: a RawNet-based model that operates directly on the raw waveform, and a 2D CNN baseline that operates on Mel-spectrogram representations. In parallel, we adopt a streamlined CLAD-style training strategy for the RawNet model, focusing on end-to-end supervised learning with robustness-oriented augmentation rather than multi-stage contrastive pre-training.

Our work is inspired by the CLAD detector, But the training pipeline is intentionally simplified. In the original framework, the encoder is first pre-trained for many epochs using a MoCo-style contrastive objective with dual encoders, a large negative-sample queue, and an auxiliary length loss, and only then fine-tuned in a separate downstream stage. In contrast, our implementation collapses these stages into a *single* end-to-end supervised training phase: the RawNet encoder and the linear classifier are optimized jointly with a standard cross-entropy loss on manipulated raw waveforms. This removes the need for a momentum encoder, a memory queue, and a long pre-training schedule, substantially reducing implementation complexity, training time, and hardware requirements, while preserving the core idea of learning manipulation-robust representations from augmented audio.

3.1 Comparative Architectures and Feature Representation

We implemented and compared two models for the binary classification task (Real vs. Fake): the RawNet encoder, serving as the robust raw-waveform detector, and a 2D CNN, serving as the conventional spectral-feature baseline.

3.1.1 The Robust Detector: RawNet Encoder ($\mathbf{D}_{\text{RawNet}}$)

The RawNet encoder is designed for end-to-end processing of the raw audio signal, and in our framework it is trained to learn manipulation-robust features directly from the waveform.

- **Feature Extractor:** We employ the `RawNetEncoderBaseline` architecture, consisting of a stack of 1D convolutional layers, followed by a three-layer Gated Recurrent Unit (GRU) and an attention mechanism. The encoder outputs a **1024**-dimensional embedding vector, denoted by $\mathbf{D}_{\text{CLAD}}$.

- **Input Preprocessing:** Each utterance is resampled to **16 kHz** and trimmed or padded to a fixed length of **64,600** samples (~ 4.03 seconds). The waveform is first normalized to a target loudness of **−23.0 LUFS** and then subjected to per-utterance z-score normalization (zero mean, unit variance).

3.1.2 The Baseline Detector: 2D CNN (D_{CNN})

The baseline detector uses a conventional time–frequency representation, relying on spectral image-like features.

- **Feature Extractor:** A lightweight **2D** Convolutional Neural Network (**SpectrogramCNN**) with three **Conv-ReLU-MaxPool** blocks, followed by a fully connected classification head that maps the spectral features to a binary real/fake prediction.
- **Input Preprocessing:** Each audio waveform is converted to a log-energy Mel-spectrogram using the following parameters: $N_{\text{FFT}} = 400$, hop length = **160**, and number of Mel bands $N_{\text{Mels}} = 128$. This results in a feature map of size **128 × 404**. Let $\text{Mel}(\mathbf{x}_{\text{raw}})$ denote the power Mel-spectrogram of the input waveform. The final input to the CNN is the stabilized log-energy,

$$\mathbf{S} = \log (\text{Mel}(\mathbf{x}_{\text{raw}}) + 10^{-6}),$$

where the small constant 10^{-6} ensures numerical stability.

3.2 Training Strategies and Experimental Pipelines

To isolate the effect of architectural choice and training strategy on robustness, we define two primary training pipelines. Unless otherwise stated, all models are trained for **10** epochs using Cross-Entropy loss ($\mathcal{L}_{\text{CE}}$) as the main objective.

3.2.1 Pipeline 1: Non-Robust Baseline (Clean Spectrogram CNN)

This pipeline serves as the conventional baseline, quantifying how a standard spectral model degrades when evaluated under out-of-distribution (OOD) manipulations.

- **Model:** D_{CNN} (**SpectrogramCNN**), as described in Section 3.1.1.
- **Training Data:** Clean log-Mel spectrograms only (**augmentation_mode** = **'none'**), i.e., no robustness-oriented augmentations are applied during training.
- **Optimization:** AdamW optimizer with a fixed learning rate of **LR** = 1×10^{-4} . The model is trained with mini-batch stochastic gradient descent on the speaker-disjoint training split.

3.2.2 Pipeline 2: CLAD-Style Robust Training (RawNet on Augmented Data)

This pipeline evaluates whether a RawNet-based representation, combined with explicit training against manipulations, yields a more robust detector.

- **Model:** D_{RawNet} (`RawNetEncoderBaseline`), operating directly on normalized raw waveforms and producing 1024-dimensional embeddings.
- **Training Data:** Robustly augmented raw audio (`augmentation_mode = 'robust_train'`), as detailed in Section 2.2. For each training sample, two manipulations are sampled at random from the set {White Noise, Volume Scaling, Fading} and applied on-the-fly to the waveform. This creates a continuously shifting acoustic domain while preserving the underlying real/fake label.
- **Optimization and Regularization:** The most performant configuration uses the AdamW optimizer with an **optimized learning rate** of $\text{LR} \approx 5.22 \times 10^{-4}$, selected via an LR-range test. In addition, **Feature Noise Regularization** is applied to the 1024D embedding vector to improve generalization. Given a clean embedding $\mathbf{f}$, the noisy embedding is defined as

$$\mathbf{f}_{\text{noisy}} = \mathbf{f} + \mathcal{N}(0, \sigma^2), \quad \sigma = 0.05.$$

This encourages the model to learn features that are stable under small perturbations in latent space. Variants of the objective that incorporate additional terms (e.g., length loss or focal loss) are evaluated as part of the ablation study in Section 4.2.

3.3 Evaluation Protocol and Metrics

Both models are evaluated under the same strictly speaker-disjoint protocol, using two complementary test domains:

- **Clean Test Set:** In-distribution evaluation on clean, non-manipulated speech segments.
- **Attacked Test Set:** Out-of-distribution stress test, where the audio has been corrupted using the stronger, fixed manipulation pipeline described in Section 2.2. This set simulates realistic attack scenarios.

Performance is quantified using the following metrics:

- **Accuracy (ACC):** Overall proportion of correctly classified samples.
- **Fake Recall (Sensitivity):** The proportion of true deepfakes correctly identified as **Fake**. This metric is critical for minimizing the False Acceptance Rate (FAR) of malicious audio.
- **Fake Precision:** The proportion of samples predicted as **Fake** that are in fact deepfakes, reflecting the reliability of positive detections.
- **Fake F1-score:** The harmonic mean of Fake Recall and Fake Precision, summarizing the detector’s ability to both capture and trust Fake predictions.
- **Equal Error Rate (EER):** The threshold-independent operating point where the False Acceptance Rate (FAR) and False Rejection Rate (FRR) are equal. A lower EER indicates a more stable and reliable detector, regardless of the specific decision threshold chosen.

This evaluation design cleanly separates the contributions of feature representation (raw waveform vs. Mel-spectrogram) and training strategy (clean vs. robustness-oriented training), providing a solid methodological basis for the comparative results reported in Section 4.

4 Results and Improvements

In this section, we evaluate the performance of our proposed methods. We first demonstrate the superiority of our best model over the standard baseline, followed by an analysis of the internal improvements (Ablation Study), and finally, we investigate the architectural robustness to confirm the necessity of raw waveform modeling.

4.1 Main Comparative Results

We conducted a two-sided evaluation to assess both in-distribution accuracy (Clean Test Set) and robustness (Attacked Test Set - Hard). Table 1 compares our final proposed configuration (CLAD with Feature Noise) against the standard CNN baseline trained on clean data.

Table 1: Performance Summary: Baseline vs. Best Proposed Model

Model	Training Condition	Acc. Clean (%)	Acc. Attacked (%)	Robustness Drop (%)
Baseline (CNN)	Clean Only	97.40	65.58	-31.82
Proposed (CLAD+FN)	Robust + Noise	96.97	96.10	-0.87

As shown, the Baseline CNN collapses under attack, losing over 30% of its accuracy. In contrast, our proposed RawNet-based model demonstrates exceptional stability, maintaining > 96% accuracy even under severe acoustic distortions.

4.2 Model Improvements and Ablation Study

To achieve this performance, we iteratively improved the RawNet backbone. Table 2 details the impact of each component on the attacked test set. The contribution of each component is detailed below:

- **Base CLAD:** Standard Cross-Entropy loss.
- **CLAD + Length Loss (LL):** Minimizing intra-class variance.
- **CLAD + Focal Loss:** Focusing on hard examples.
- **CLAD + Optimized LR:** Fine-tuning the learning rate schedule.
- **CLAD + Feature Noise (FN):** Injecting Gaussian noise into latent embeddings.

Table 2: Ablation Study: Impact of Improvements on Attacked Data

Configuration	Recall (Fake)	Precision (Fake)	Accuracy
Base CLAD/RawNet	79.22%	96.83%	88.31%
CLAD + Focal Loss	87.01%	93.06%	90.26%
CLAD + Length Loss	88.31%	93.15%	90.91%
CLAD + Optimized LR	96.54%	93.31%	94.81%
CLAD + Feature Noise	95.67%	96.51%	96.10%

4.3 Latent Space Visualization

To visualize the evolution of robustness, we projected the embeddings of the Attacked Test Set into 2D space using t-SNE. Figure 1 displays the feature space of the Baseline CNN followed by the progressive improvements of our CLAD models.

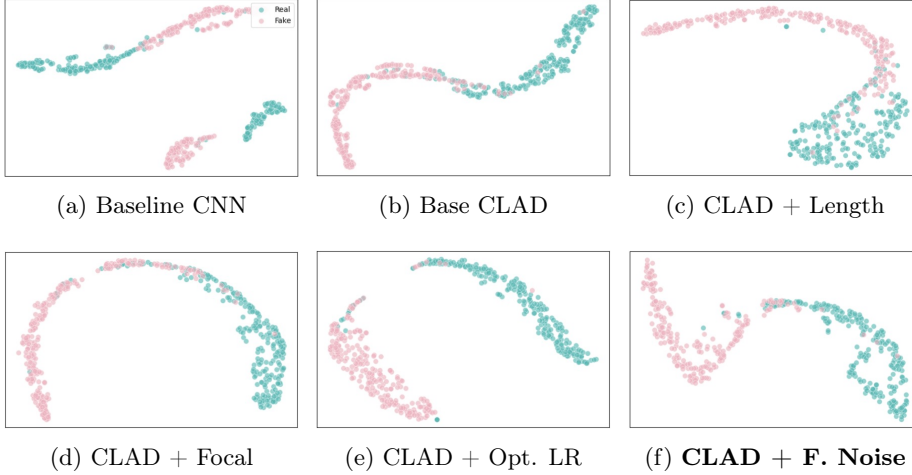


Figure 1: t-SNE Visualization under attack. (a) The Baseline CNN shows complete class overlap. (b-e) The CLAD models show gradual improvement in separation. (f) The final model with Feature Noise exhibits distinct, well-separated clusters, explaining its superior 96.10% accuracy.

4.4 Architectural Robustness: Raw Waveform vs. Mel-Spectrogram

To verify whether our results stem from the raw waveform architecture or merely from the data augmentation, we trained a secondary CNN instance (“Robust CNN”) on the exact same augmented data pipeline used for CLAD.

Table 3: Architectural Comparison: Accuracy vs. EER on Hard Attacks

Model Architecture	Training Data	Accuracy (%)	EER (Lower is Better)
Spectrogram (CNN)	Clean Only	65.58%	$\sim 28.0\%$
Spectrogram (CNN)	Robust (Augmented)	95.02%	7.36%
Raw Waveform (Ours)	Robust (Augmented)	96.10%	6.06%

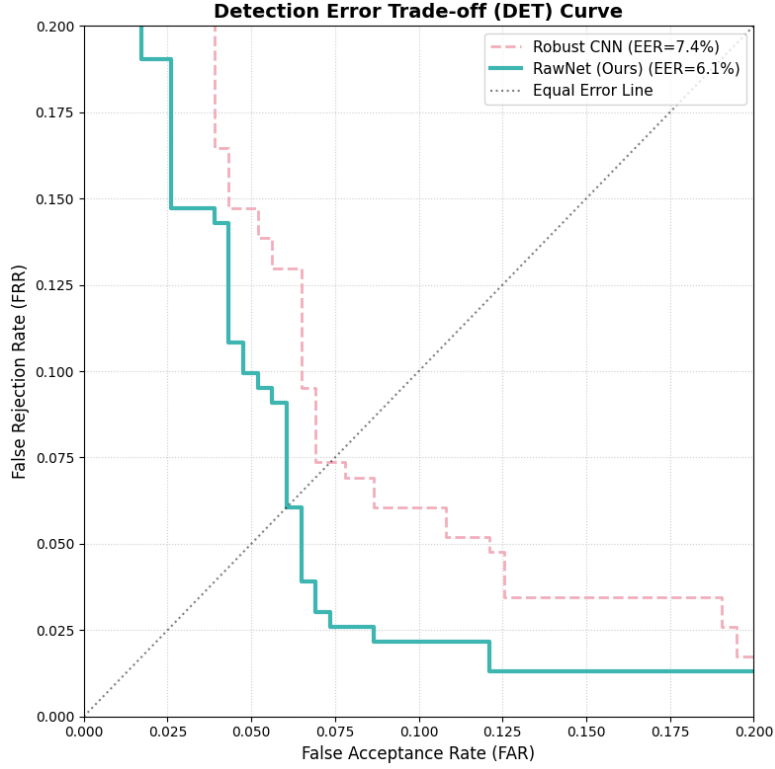


Figure 2: **Detection Error Trade-off (DET) Curve.** The RawNet model (Blue) maintains a consistently lower error rate than the Robust CNN (Red) across different decision thresholds, highlighting its superior stability for security applications.

Analysis: The results reveal that data augmentation is a critical driver of robustness, boosting the CNN’s accuracy to 95.02%. However, the **RawNet architecture demonstrates superior stability.**

While accuracy scores are comparable, the RawNet model achieved a significantly lower Equal Error Rate (EER) of **6.06%** compared to the CNN’s 7.36%. This represents a **relative error reduction of $\approx 17.6\%$** . As shown in Figure 2, the RawNet curve consistently stays below the CNN curve, confirming that the raw-waveform approach provides a more reliable boundary between real and fake audio, regardless of the chosen operating threshold.

5 Conclusion and Future Work

This project presented a CLAD-inspired, RawNet-based framework for robust Audio Deepfake Detection under realistic manipulation attacks. We constructed a speaker-disjoint benchmark based on genuine VCTK recordings and XTTS-generated deepfakes, and evaluated two competing architectures under a shared protocol: a RawNet encoder operating directly on normalized raw waveforms, and a 2D CNN baseline operating on log-Mel spectrograms.

Unlike the original CLAD framework, which relies on a heavy two-stage MoCo-style contrastive pre-training with dual encoders, a memory queue, and an auxiliary length loss, our approach adopts a single-stage, end-to-end supervised training pipeline. The RawNet encoder and its classifier are optimized jointly with a standard cross-entropy objective on aggressively augmented raw audio, while the spectrogram-based CNN is trained on clean Mel-spectrograms. This design preserves the core idea of robust learning representations based on augmented data manipulation, but substantially reduces the complexity of implementation, training time, and hardware requirements.

Empirically, our results highlight the dominance of a data-centric training strategy. Subjecting the lightweight CNN to our augmentation pipeline boosted its accuracy from 61.9% to 95.02%, proving that simple models can be robust given the right data.

However, the **RawNet-based detector remains the superior solution for deployment**. While accuracy scores were comparable, the RawNet model demonstrated greater reliability with an **Equal Error Rate (EER) of 6.06%**, significantly outperforming the CNN’s 7.36%. Additionally, by removing the spectrogram generation step, the RawNet pipeline offers a streamlined, end-to-end defense mechanism that eliminates preprocessing latency, making it highly suitable for real-time voice authentication systems.

Future Work: Several directions remain open for further investigation:

- **Unseen Synthesis and Cross-Corpus Generalization:** Extending the evaluation to additional TTS/VC systems, datasets, and recording conditions in order to better quantify the model’s ability to generalize to unseen generative models and acoustic domains.
- **Richer and Adversarial Attack Modeling:** Incorporating a broader family of manipulations, including room acoustics, codec distortion, dynamic range compression, and adversarial perturbations. This would enable a more exhaustive stress test of robustness in deployment-like scenarios.
- **Advanced Training Objectives and Pre-training:** Exploring lightweight forms of self-supervised or contrastive pre-training, as well as alternative loss functions, with the goal of further improving robustness without reintroducing the full complexity of the original CLAD pipeline.
- **Scalability and Deployment:** Adapting the proposed RawNet-based detector to low-latency, streaming scenarios (e.g., real-time call monitor-

ing), and studying calibration and thresholding strategies to control false acceptance and rejection rates under operational constraints.

Overall, this work demonstrates that a simplified, end-to-end RawNet architecture, combined with a principled robustness-oriented training strategy, can provide an effective and computationally efficient foundation for next-generation Audio Deepfake Detection systems.